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Toward a theory of agency and altruism in family firms

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Abstract

Surprisingly, the majority of U.S. family firms offer employed family members short- and long-term performance-based incentive pay. We draw on the household economics and altruism literatures to explain why family firms might feel compelled to do so and develop theory that predicts when this practice will be beneficial. Results based on data obtained from 883 family firms show that altruism, as reflected by the parents' estate and share transfer intentions, moderates the effect of these pay incentives as our theory predicts. As such, this article helps explain both how altruism influences agency relationships in family firms and why business practice in family firms differs from those found in other types of firms.

Keywords: Family firms; Altruism; Agency

1. Executive summary

Contrary to the tenets of agency theory, the majority of U.S. family firms offer employed family members short- and long-term performance-based incentive pay. We draw on the household economics and altruism literatures to explain why so many family firms do so, and develop theory that predicts when this practice will be beneficial. Our thesis is that agency relationships in family firms are distinctive because they are embedded in the parent–child relationships found in the household, and so are characterized by altruism. Altruism compels

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parents to care for their children, encourages family members to be considerate of one another, and fosters loyalty and commitment to the family and firm. Altruism, however, has a dark side in that it can give both parents and children incentive to take actions that can threaten the welfare of the family and firm alike. As we will show, integrating both the bright and dark sides of altruism into the standard agency model accommodates that model to this domain. To this end, we then use survey data from 883 family firms and show that altruism, as reflected by the parents' estate and share transfer intentions, moderates the effect of these pay incentives as indicated by our theory. Contrary to the prediction of agency models that do not account for altruism, our results demonstrate that the family ownership does not appear to represent the kind of governance panacea that is often attributed to owner-management by agency theorists. Rather, and consistent with the case-based studies about family firms, family-owned and -managed firms appear to experience agency problems that can be very costly to mitigate. Our results also suggest how altruism influences agency relationships in family firms and helps explain why the conduct of these firm often differs from that demonstrated by other types of business organizations.

2. Introduction

Family-owned and -managed firm's play a vital role in the United States and world economies, yet the relevance of existing theory to this form of governance has yet to be evaluated and tested. Agency theory, for example, posits that family owner-management promotes communication and cooperation within the firm and guards against opportunism, sparing them the need to closely monitor management or the expense of pay incentives. However, evidence about family-owned and -managed firms (henceforth, family firms) is at odds with this conclusion. Levinson (1971, p. 90) suggests that family firms are "[. . .]plagued by conflicts" that cause many to founder, and survey data indicate that over 70% of them offer pay incentives to employed family members (Fraser, 1990; Greco, 1997).

The purpose of this article is to reconcile this gap between theory and evidence. Our thesis is that agency relationships in family firms are distinctive because they are embedded in the parent–child relationships found in the household, and so are characterized by altruism. Altruism is powerful force within family life and, by extension, within the family firm. It compels parents to care for their children, encourages family members to be considerate of one another, and fosters loyalty and commitment to the family and firm. Altruism, however, has a dark side in that it can give both parents and children incentive to take actions that can threaten the welfare of the family and firm alike. As we will show, integrating both the bright and dark sides of altruism into the standard agency model accommodates that model to this domain, and allows it to explain theoretically curious phenomena, like why so many family firms offer family agents pay incentives.

We begin by reviewing the salient points from the economic theory of the household and the altruism literatures and use those theories to generate agency propositions that are relevant to family firms. We then identify circumstances when altruism's effects might be observed, and test hypotheses using data from a sample of family-owned and -managed corporations obtained from a large-scale survey conducted by Arthur Andersen. We conclude with a discussion of the implications of our research for theory and practice.

3. Background: altruism and agency in households

Religious and research perspectives on altruism differ. Theologians tend to view altruism as a moral value that motivates individuals to undertake actions that benefit others without any expectation of external reward (Batson, 1990). Altruism is therefore external or exogenous to man's fundamental character, acquired through moral education, and viewed as a manifestation of God's grace and evidence of faith. Sociologists, in contrast, tend to view altruism as a trait or preference that is endogenous to man's character and based, at least in part, on feelings, instincts, and sentiments (Lunati, 1997). Economists hold a similar view and generally model altruism as a utility function in which the welfare of one individual is positively linked to the welfare of others (Bergstrom, 1989). The incentive it provides is therefore powerful and self-reinforcing because efforts to maximize one's own utility allow the individual to simultaneously satisfy both altruistic (other-regarding) and egoistic (self-regarding) preferences. Parents, it follows, are not only generous and charitable to their children because they love them but also because their own welfare would decline if they acted in any other way (Becker, 1981).

Simon (1993) and Eshel et al. (1998) note that altruism compels parents to care for their children, encourages family members to be considerate of one another, and makes family membership valuable in ways that both promote and sustain the bond among them. This bond, in turn, transfers a history, identity, and language to the family that makes each unique. The intimate knowledge among family members facilitates communication and decision-making (Gersick et al., 1997). Finally, altruism also fosters loyalty and commitment to the family and to its prosperity (Ward, 1987).

Altruism, however, can create agency problems within the household. For example, since altruism is at least partly motivated by the parents' desire to enhance their own welfare, parents have incentive to be generous although that generosity may cause their child to free ride (e.g., leave an assigned household chore for a parent to complete), shirk (e.g., squander their parent's money), and/or remain dependent upon their parents. Parents are thus faced with a "Samaritan's dilemma" (Buchanan, 1975) in which their actions give beneficiaries incentive to take actions or make decisions that may ultimately harm their own welfare.

More broadly, the Samaritan's dilemma is representative of class of agency problems associated with the exercise (or lack) of self-control by the principal. Self-control problems arise whenever parties to a contract have both the incentive and ability to take actions that "harm themselves and those around them" (Jensen, 1994; 43). While many self-control problems are trivial (e.g., cheating on a diet), agency problems arise when a loss of self-control causes a principal to violate the terms, or perhaps even the spirit, of an existing contract or agreement. Parents recognize the Samaritan's dilemma and attempt to deal with it through self-discipline and education (e.g., exercising moderation and/or framing their children's expectations), preemption (e.g., implementing an estate plan), or contingent contracts (e.g., tied transfer agreements) (O'Donoghue and Rabin, 2000). Tied transfer agreements, for example, mitigate the agency threat by making the transfer of a promised benefit contingent on their children's behavior (Lindbeck and Weibull, 1988). Tied transfer schemes are ubiquitous and can be crafted to fit virtually any desired period, including the short (''You get your allowance only after your chores are done''), near (''Do well in high school and I'll pay for your college education''), and long term (''Someday this might all be yours!'').

Parents like tied transfers because they seemingly promise that parents can continue to be generous without spoiling their children or harming relationships they have each other or with their parents. The hope that these schemes can be effective, however, rests on two assumptions: that the children will recognize that they will be better off if they comply with the altruist's wishes and that altruistic parents can (and will) take actions that prevent their children from bettering their own welfare at the cost of another family member. For example, theory presumes that because an altruistic parent's welfare will also fall if one child's actions harm another's, parents have incentive to restore equity within the household, either by making transfers that undo the damage done or changing the allocation of the household ''pie'' (Becker, 1981).

Bergstrom (1989), however, demonstrated that these assumptions are false. First, the agency problems that parents seek to remedy often arise because children can manipulate indulgent parents by working too little or spending too much. This manipulation creates information asymmetries that make it difficult for parents to identify transfer plans that will have the desired effect on their children's conduct. These asymmetries also increase the odds that any plan that parents might offer their children will be inefficient in the sense that it is either larger or smaller than what is needed to motivate their children and/or is inconsistent with their child's preferences.

A second reason tied transfer plans fail is that children recognize that altruism can make it difficult for parents to enforce their plans. Children learn quite early whether they can persuade their parents to relax the criteria to which the transfer was originally tied. Children also learn from experience whether changed circumstances, such as those that threaten family welfare, will cause parents to unilaterally alter, or even abandon, existing agreements. The need to care for a family member's health is an example, as is aging, which is strongly associated with a decline in the parent's marginal propensity to expend resources on their children (Bergstrom, 1989, p. 1140). As Thaler and Shefrin (1981) point out, the fact that principals may have incentive to unilaterally alter the terms of existing agreements increases agency costs, by giving agents incentive to invest resources in monitoring the principal and in attempts to influence their conduct. Not coincidentally, Bergstrom's ''Case of the Prodigal Son'' is itself modeled after the biblical parable of the same name that describes the consequences of an unexpected event that caused a parent to violate the perceived spirit of an existing tied transfer agreement.

Lastly, tied transfer plans may not produce the desired outcomes because it is presumed that parents are capable of distributing resources within the household in a manner that optimizes family welfare. But parents are capable of doing this if conditional transferability exists; that is, if each child's tastes or preferences for different goods can be expressed in terms of a single commodity, like money. Put differently, Bergstrom (1989) argues that parents will only be able to precisely balance one child's taste for goods like leisure with another child's taste for goods like money, power, or status if "money is enough." Unfortunately, money is never enough because tastes and utilities differ among individuals. Worse, altruism biases and filters the information parent's process about their children ("Junior would *never* do that!") and can cause even well-intended children to provide misleading information about their taste ("I just loved your gift!"). These information asymmetries make it difficult for altruistic parents to be both generous and just and increases the risk that tied transfer plans may not only fail to properly motivate the beneficiary, but could also spark conflict or jealousy among siblings and within the household. In the next section, we posit that altruism has a similar effect on agency threats and costs in family firms.

3.1. Altruism and agency in the family firm

Our thesis is that the nature of agency relationships in family firms is embedded (i.e., influenced, but not fully determined) by the past and ongoing parent–child relationships of the household, and therefore is characterized by altruism. Altruism makes each family member employed by the family firm a de facto owner of the firm in the sense that each acts in the belief that they have a residual claim on the family's estate (Stark and Falk, 1998). According to agency theory, ownership should align the interests among family agents toward growth opportunities and risk thereby reducing the cost of reaching, monitoring, and enforcing agreements. Thus, there should be little need to monitor family agent performance. Altruism should also increase communication and cooperation within the family firm thereby reducing information asymmetries among family agents and increase their use of informal agreements (Daily and Dollinger, 1992). Finally, altruism should create a heightened sense of interdependence among family agents, since employment links their welfare directly to firm performance. In short, the agency benefits of altruism should be pronounced.

We deduce, however, that agency problems rooted in altruism and self-control are exacerbated when the privileges of ownership place control of the firm's resources at the CEO's discretion. This broadens the CEO's capacity to make altruistic transfers (like employment, perquisites, and privileges), and entitles family members to benefits that would not be received were they employed elsewhere. These privileges, and the sense of entitlement they often evoke (Gersick et al., 1997), can create a variety of agency costs. First, they can increase the need to monitor the work of family agents because their hire is determined by family status as well as by their professional qualifications. Put differently, because "important management decisions are made based on wealth and willingness to bear risk, as well as decision skills" (Fama and Jensen, 1983, p. 332), altruism exposes family firms to adverse selection–the agency threat associated with lack of ability as opposed to lack of effort. Monitoring is therefore required to assure that the decisions and activities undertaken are appropriate and commensurate with the family agent's position and level of authority. The risk that the perquisites and privileges granted by the CEO may spark a "Samaritan's dilemma" can also make close supervision necessary.

Second, altruism reduces the CEO's ability to effectively monitor and discipline family agents. Just as it was in the household, altruism systematically biases the CEO's perceptions, and hence the information that they filter and process about employed children. Moreover, altruism reduces the effectiveness of a CEO's supervision because family agents, like they were when they were children, tend to free ride on the CEO whenever the responsibilities of the CEO and family agent overlap (Lindbeck and Weibull, 1988). Finally, the CEO's ability to discipline family agents is compromised by both the CEO's altruism, and by the ramifications that such actions might have on familial relationships inside the firm and among the extended family outside the firm (Donovan, 1995).

In summary, agency theory suggests that because ownership aligns interests among family agents, family firms should have a little need to incur agency costs such as those associated with monitoring and pay incentives. We deduce a different proposition: The desire to maximize family welfare can compel family firm CEOs to take actions that create positive agency costs. However, because altruism compromises the CEO's ability to monitor and discipline family agents, we anticipate that there is a positive relationship between the families' welfare and the use of pay incentives with family agents.

3.2. Pay incentives and family agents

Pay incentives are tied transfers that make a portion of an agent's pay contingent upon some performance objective, usually firm performance (Andersen, 1985). Their purpose is to shift risk from the principal to the agent thereby removing the agent's motivation to keep information private and/or take actions that are harmful to the principal's interests. Agency theory suggests that it should not be necessary to offer pay incentives to family agents because they, as de facto owners, already have their personal wealth tied closely to the value of the firm. Yet, surveys find that between 73% and 85% of all family firms offer cash bonuses (Fraser, 1990; Greco, 1997).

Will pay incentives like bonuses have the desired positive performance outcomes in family firms? The answer is neither simple nor straightforward. First, for a tied transfer like a pay incentive to be effective, the recipient must perceive that its value is commensurate with the effort required to earn it. However, the relative value of any incentive is contingent on the value of other unearned transfers, or entitlements, the family agent expect to receive. Moreover, the value of all transfers is clouded by uncertainty as to whom, how much, and when the CEO will transfer ownership and control of the firm. Thus, the fact that a pay incentive is earned, while other transfers are not, and the fact that uncertainty gives family agents reason to steeply discount the value of any transfer payment, makes it difficult for both the CEO and the family agent to properly "price" the incentive.

On the surface, the latter point might not appear to apply to cash bonuses, whose value should be more transparent to family agents due to their time horizon. Embedded within both long- and short-term tied transfers at family firms, however, is an implicit disincentive. This disincentive emanates from the fact that the share of the marginal wealth family agents generate from their own industriousness, and to which they as an owner feel entitled, may not be paid to them in proportion to how and/or when it was earned. Rather, they know that at least part of these payments are at risk because the CEO has incentive to let a variety of factors associated with altruism and ownership (e.g., the financial needs of the firm or of other family members) influence their award (Bergstrom, 1989).

It follows that altruism can prevent tied transfers from having the desired performance outcomes if family agents face too much uncertainty to adequately calibrate the value of their effort. Extending this inference, we posit that if principal–agent relationships in family firms are indeed characterized by altruism, then the effect of pay incentives on family agents should vary with what they know about the CEO's transfer intentions. In the following sections, we identify three circumstances under which we think different information about the CEO's transfer intentions could cause the relationship between pay incentives and firm performance to change.

3.2.1. The effect of selling the firm

We infer four reasons from the altruism literature that lead us to expect that pay incentives will be effective when CEOs declare their intentions to sell their firms but not when family agents expect that the firm will remain under family control.

First, the expectation that the firm will be sold reduces information asymmetries because family agents then have less incentive to compete with each other, to squander resources, and to seek the CEO's favor (Buchanan, 1983). Also, the expectation that the firm will be sold would make all the family agents' claims to rival goods, like the status and authority that accompany ownership and control of the firm, irrelevant. Second, information asymmetries should fall because family agents have less reason to be concerned about receiving an equitable proportion of the altruist's estate when it is comprised principally of money, which is homogeneous and easy to distribute equitably, as opposed to when it is comprised of ownership in the firm, whose relative value varies both with circumstance and amount of influence family agents exercise in the management of the firm. Put differently, stating the intention to sell improves the transferability of utilities among the family agents, while making information flows more certain and symmetric. Third, reduced information asymmetries improve the ability of the CEO to craft effective incentive plans, as will the fact that the impending sale shortens the time period over which the family agents discount the value of any incentive. Finally, the anticipated sale can liberate family members whose altruistic feelings toward the family might have compelled them to work in the firm (Gersick et al., 1997; Lansberg, 1983; Nelton, 1995) and lowers exit costs by causing family members to exclude the value of the perquisites and privileges that accompanied employment in the family firm from their future earning calculations. Since agency theory indicates that agents become less risk averse as exit costs fall, incentives should become more effective as the expectation that the family firm will be sold reduces the cost of shifting risk from the CEO to the family agent (Hill and Jones, 1992). Collectively, these arguments lead us to predict:

H1: There is a positive relationship between firm performance and the use of pay incentives with family agents when those agents anticipate that the firm will be sold, but no relationship when they anticipate that the firm will remain in family hands.

3.2.2. The effect of transfer intentions

Similarly, we infer from the research literature that the effectiveness of pay incentives in firms that will not be sold depends on how much family agents know about the CEO's estate and share transfer intentions. Specifically, we posit that *disclosure* of the CEO's estate and share transfer plans moderates the relationship between pay incentives and family firm performance. Recall that altruism grants family agents de facto ownership, and with it, a belief that they have the right to claim a share of the marginal wealth generated by their effort. A CEO's public commitment to an estate and share transfer plan reduces uncertainty regarding the value of that right. All else being the same, this information should help family agents accurately calibrate the value of their efforts and thereby improve the motivational effect of a pay incentive. Disclosure of the CEO's estate and transfer plans should also help reduce the amount of rivalry among family agents, which in turn should reduce information asymmetries. Simply put, family agents have little reason to continue lobbying for a promotion, added compensation, or an increased share of the estate, once the CEO's intentions are made public.

Arguments for disclosure assume that family firm CEOs will intuit the agency implications of short- and long-term pay incentives, just as many parents intuit the agency implications of the Samaritan's dilemma, and therefore decide to make the specifics of their estate plans known. However, family firm CEOs have a variety of reasons to keep this information private. For example, they may feel they need added time to assess the capabilities of different family members or fear that disclosing how the family shares will be distributed will cause a jealous rift within the family, as what happened in the biblical story about the sons of Jacob who, upon learning that their brother Joseph had been chosen to inherit the family birthright, sold him into slavery. Perhaps the firm employs spouses, cousins, or other members of the extended family. This not only complicates the drafting of the estate plan, but also virtually assures that some family agents will feel slighted, and perhaps even disenfranchised, once the estate plan is revealed. Indeed, the annals of family business are filled with examples of firms that were torn apart by the ensuing conflict (Donovan, 1995; Levinson, 1971).

The result is that family firm CEOs face a peculiar dilemma: The information required to make pay incentives more effective—and thereby solve one set of altruism-induced agency problems—may expose the firm to another set of agency problems. Thus, our view is that the effectiveness of pay incentives in those firms that will remain under family control is contingent on whether CEOs publicly commit their estate and share transfer intentions. Our view also sheds light upon why some family firm CEOs may decide to keep their transfer intentions private, and thus, perhaps unintentionally undermine the effectiveness of those incentives. It follows that:

H2: There will be a positive relationship between firm performance and the use of pay incentives with family agents when those agents expect that the firm will remain in family hands if their estate and share transfer plans are known, and no relationship if these plans are not known.

3.2.3. The timing of the transfer

Our view of altruism and agency in family firms also sheds light on why even those CEOs who have declared their transfer intentions may delay implementation of a transfer plan or postpone retirement (Handler, 1990). First, concern for the welfare of the family as a whole may cause some CEOs to make decisions that compromise the welfare of some family agents. For example, some CEOs delay retirement because they feel that their chosen successor had not yet attained the skills and experience needed to lead the firm (Handler, 1990). Other CEOs might put off the transfer of shares or delay retirement until they have secured [or ensured] their "nest egg" (Rubenson and Gupta, 1996). Information asymmetries exacerbate these aforementioned risks (Handler, 1990; Sonnenfeld, 1988). Whatever the reason for delay, the uncertainty that it causes family agents should act as a disincentive. It follows then that the more uncertain the date of the CEOs retirement (and hence the timing of the transfer), the more they will discount the value of their pay incentive. This raises the prospect that:

H3: There will be a positive relationship between firm performance and the use of pay incentives with family agents for firms that are expected to remain in family hands and where the estate and share transfer intentions are known if the expected date of retirement is near and no relationship if the expected date of retirement is distant.

4. Data, measures, and method

4.1. Data

Reliable information on family firms is extremely difficult to obtain (Handler, 1989). The availability of data from a 1995 survey of American family businesses that had been designed and administered by The Arthur Andersen Center for Family Business was therefore serendipitous. This survey is one of the largest and most comprehensive surveys conducted on family firms (Gersick et al., 1997). Andersen statisticians assure us that it is reliable and representative of the population they set out to study.

The survey instrument was reviewed by a focus group of family business owners and pilot tested on a holdout sample, before being mailed to the chief executive in 37,500 privately held U.S. family businesses. Consistent with other large-scale surveys administered by professional consulting firms, this single mailing yielded 3860 responses within 1 month, or a response rate of 10.3%. This rate is comparable to "the 10–12 percent rate typical for studies which target executives in upper echelons" (Geletkanycz, 1998, p. 622; Koch and McGrath, 1996) and chief executives in small- to medium-sized enterprises (MacDougall and Robinson, 1990).

Since the Andersen survey (Arthur Andersen & Co., 1995) was designed to obtain information from all types of family businesses, ranging from "mom-and-pop" proprietorships to large professionally managed family-controlled corporations, we applied a number of ex post screening criteria to narrow the list of responding firms. First, we deleted 1766 partnerships, proprietorships and S corporations because different laws and tax policies influence both share transfer and compensation practice at these firms. Another 539 cases were also dropped due, principally, to missing ownership data. Lastly, we deleted 627 firms that had less than US$5 million in sales. Our sample thus excludes "lifestyle firms," i.e., small firms that might be operated mainly for the purpose of income substitution (Allen and Panian, 1982), as well as other firms for whom growth is not a strategic objective (Carland et al., 1984; Rubenson and Gupta, 1996). Larger firms are less likely to be operated in this manner since the demands of managing them mitigate the family/CEO's primary motive for suppressing growth; to more easily maintain managerial and ownership control (Daily and Dollinger, 1992; Whisler, 1988). Our final sample consisted of 883 relatively large, established, family-owned and -controlled corporations. The average firm in this subsample of family firms has annual sales of US$38 million, 181 employees, and is 51 years old.

4.2. Dependent variable

Consistent with Jensen and Meckling's (1976, p. 313) assertion that agency relationships within the firm have a strong bearing on its growth rate, family firm performance is defined in terms of "sales growth." Sales growth is considered to be a more reliable dimension of family firm performance than traditional income-based measures because privately held firms have incentive to minimize reported taxable income and no incentive to minimize reported sales (Dess and Robinson, 1984). Like Cavusgil and Zou (1994), the Andersen survey measured sales growth as a 5-year average, using a six-point, nominal scale ranging from decreased, no change, increased 1–5%, increased 6–10%, increased 11–20%, to increased 21% or more. The measure is self-reported, as must be the case with studies of privately held firms. However, self-reported performance measures by executives have been shown to be reliable (Tan and Litschert, 1994), particularly when reported on anonymous surveys (Dillman, 1978).

4.3. Independent variables

Family incentives is a dummy variable (1/0) that we used to identify firms from the survey that offer pay incentives in the form of an annual and/or long-term cash bonus to family members. In total, 71% of 883 firms in our sample indicated that they offer pay incentives to family members, while 69% offered the same to nonfamily agents, figures fully consistent with the range reported in four other surveys (previously cited). Family firms "rely heavily on cash incentives" (Fraser, 1990, p. 58), and not stock incentives, due to both a concern for maintaining control of the firm, and the fact that cash holds a greater incentive than stock for employees when the firm's equity is not publicly traded (Berk et al., 1988).

To assure that any observed relationship between pay incentives and the dependent variable is driven by cash bonuses paid to family agents, we control for nonfamily pay incentives by including this variable as a covariate, coded (1/0) like family pay incentives. To assure that the cash bonus paid to family agents is indeed a contingent portion of salary and not simply a dividend payment that is distributed as a salary bonus to reduce the tax burden on family members, we identify firms that pay dividends with a dummy variable and also include it as a covariate in the regression.

4.4. Moderators

We used dummy variables to investigate the contingent influence of the estate and share transfer intentions on pay incentives. First, we used an item that asked "Who do you anticipate will be the next owner of the firm?" to distinguish the 303 firms in our sample that anticipate the firm will be "sold to outside investors" from the 580 firms that intend to "transfer to family." We measured estate plans with the item "do the significant shareholders of the business know each others' estate plans and share transfer intentions?" to distinguish the 296 firms where the share transfer intentions remain unknown from the 587 firms where the share transfer intentions are known. Finally, we measured "executive tenure" from the item "when does the CEO plan to retire from this position" to distinguish the 417 firms whose CEO intends to retire within the next 5 years from the 466 whose CEOs have no such intention.

4.5. Covariates

We included 11 covariates to reduce variance that is extraneous to the research question or may confound interpretation of our study. This list includes the two variables previously mentioned, nonfamily pay incentives and dividends, as well as firm size, firm age, export sales, technological intensity, capital intensity, industry growth, multiple family ownership, number of employed family members, and percentage of family ownership. Each is described in Appendix A.

5. Results

Descriptive statistics and correlations are reported in Table 1. The results of moderated hierarchical regressions for H1 to H3 are reported in Table 2. In all cases, the F statistic associated with the set of covariates is significant at the .05 level or better thereby attesting to the efficacy of the covariate function. The F statistic associated with entry of the hypothesized variables after adjusting for the main effects is significant at the .05 level or better for H1 and H2 and significant at the .07 level for H3. To facilitate presentation, only the results for the main and higher order effects are presented.

The regression results in column 1 of Table 2 show that the main effect of family pay incentives has a positive but nonsignificant relationship with firm performance after taking into account the main effects of all variables, including the effect of nonfamily pay incentives, dividend, estate plans, and anticipated time to CEO retirement. Recall that this finding is consistent with the pay practices of about 80% of family-owned and -managed firms, but inconsistent with agency theoretic models that do not incorporate the effects of altruism. However, the subsequent analyses of H1 to H3 suggest that this nonsignificant main effect is moderated by the predicted factors.

For example, Table 2 shows that the influence of pay incentives on family firm performance, net of the influence of 11 covariates, is moderated ($P<.001$) by whether

Table 1
Descriptive statistics

Variables	Mean	S.D.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Sales growth	3.86	1.44															
2. Sales revenues	1.22	0.43	.10														
3. Company age	50.55	27.85	−.12	.14													
4. Export sales	1.48	0.81	.08	.12	.03												
5. Information technology	3.07	0.91	.11	.18	.07	.07											
6. Industry sales growth	0.55	0.50	.10	.04	.12	.25	−.01										
7. Asset intensity	0.68	0.47	.08	.07	.10	.22	.02	.54									
8. Multiple family ownership	1.16	0.53	−.01	.02	.01	.01	−.03	.01	.02								
9. Number of family employees	3.48	2.18	.08	.11	.02	−.02	.00	.04	.03	.06							
10. Family ownership (%)	85.46	25.27	.05	−.15	−.16	−.10	−.01	−.08	−.06	−.11	−.04						
11. Dividends	0.32	0.47	.00	.11	.19	.07	.07	.03	.04	.06	.06	−.13					
12. Nonfamily incentive pay	0.69	0.46	.11	.09	.00	.09	.05	−.07	−.04	−.07	−.03	−.08	.08				
13. Stay in family? (intent)	0.68	0.47	−.02	−.03	.06	−.06	−.06	−.07	−.01	.03	.15	.01	.03	−.02			
14. Executive tenure (years to retirement)	2.42	1.14	−.05	−.04	−.07	−.05	−.10	.00	.01	−.03	.11	.04	−.02	−.01	.11		
15. Family incentive pay	0.71	0.45	.12	.06	−.02	.04	.06	−.07	−.04	−.01	−.01	−.01	.07	.64	.01	.01	
16. Estate plans known	0.65	0.48	.06	−.01	−.05	−.05	.03	−.11	−.04	−.09	−.02	.13	−.03	.00	.12	.01	.03

$N = 883$. Correlations larger than .05 are significant at $P < .05$.

Table 2
Moderated hierarchical regressions

Variables	Sales growth main effects	Sales growth hypotheses
1. Sales revenues	.29***	.33****
2. Company age	– .01****	– .01****
3. Export sales	.05	.06
4. Importance of information technology	.14***	.13***
5. Industry sales growth	.33****	.32****
6. Asset intensity	.09	.09
7. Multiple family ownership	– .01	– .02
8. Number of family employees	.05***	.06***
9. Family ownership (%)	.01 *	.01 *
10. Dividends	– .01	.02
11. Nonfamily incentive pay	.20	.21
12. Stay in family? (inherit)	– .01	.63**
13. Executive tenure (years to retirement)	– .08 *	– .44****
14. Family incentive pay	.22	.75****
15. Estate plans known?	.16 *	.41
16. Inherit × Family Incentive Pay		– .87****
17. Estate Plans × Family Incentive Pay		– .37
18. Inherit × Estate Plans		– .46
19. Inherit × Estate Plans × Family Incentive Pay		.75****
20. Executive Tenure × Family Incentive Pay		.38***
21. Executive Tenure × Inherit		.46**
22. Executive Tenure × Inherit × Family Incentive Pay		– .42 *
F	5.21****	4.27****
Change in F	–	2.15**
Adjusted R^2	.07	.08
N	883	883

* $P<.10$.
** $P<.05$.
*** $P<.01$.
**** $P<.001$.

respondents expect the firm to be sold or kept in family hands. Subgroup analysis (available upon request) suggests a similar conclusion: Pay incentives are not associated with firm performance in the 580 firms that are expected to be kept in family hands (inherit = 1) but are positively associated with firm performance ($P<.05$) in the 303 firms that are expected to be sold (inherit = 0). Moderation is also evident in the plot of the two-way interaction (Fig. 1), which shows that pay incentives have no apparent benefit when it is anticipated that the firm will be kept in the family, but are beneficial when this is not the case.

The data also lend support to H2. Table 2 shows that the influence of pay incentives on family firm performance is moderated ($P<.001$) by whether or not the estate and share transfer intentions of the principal shareholders are known in the sample of 580 firms that are expected to be kept in family hands. Subgroup analysis reveals that for firms that are expected

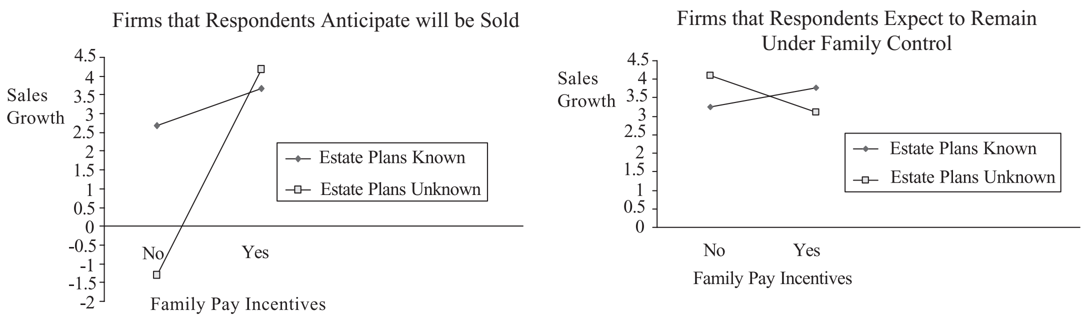

Fig. 1. The influence of knowledge about estate plans on the effectiveness of family pay incentives.

to remain in family hands, pay incentives are positively tied to firm performance ($P < .01$) in the 403 cases in which the estate and share transfer intentions are known (commitment = 1), but not in the 177 firms in which these intentions are not known (commitment = 0). Consistent with results for H1, family pay incentives are positively associated with firm performance in the 303 firms that expect to be sold, regardless of knowledge about the major shareholders estate plans. The plot of the three-way interaction in Fig. 2 both confirms these regression results and hints that pay incentives may function as a disincentive when transfer intentions remain ambiguous.

Lastly, the data marginally support H3. Table 2 shows that in firms that are expected to remain under family control, pay incentives appear to influence family firm performance ($P < .07$) when the anticipated date of the CEO's retirement is soon (within 5 years) but not distant. Subgroup analysis confirms that for firms that are expected to remain under family control, pay incentives are positively associated with firm performance ($P < .05$) for the 303 firms that expect to be kept in family hands (commitment = 1) and anticipate the CEO will retire soon (retirement = 1), but not in 277 firms in which the anticipated date of the CEO's retirement is distant (retirement = 0). The influence of the proximity of retirement on incentive compensation is apparent in the plot of H3 in Fig. 2.

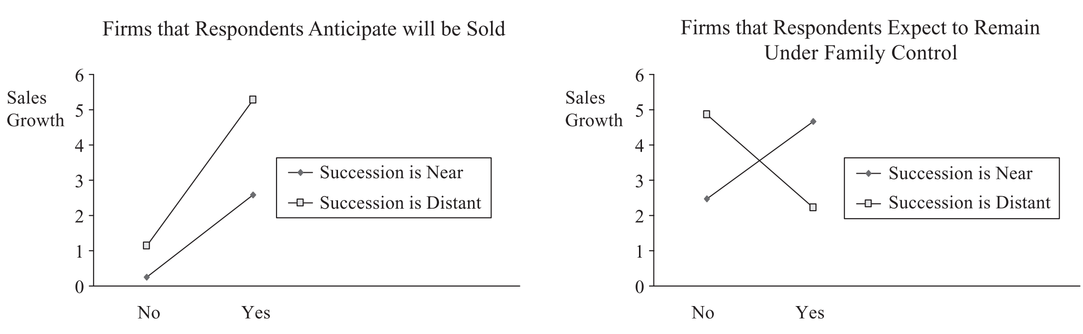

Fig. 2. The influence of approaching succession on the effectiveness of family pay incentives firms that respondents expect under family control.

6. Discussion

Simon (1993, p. 160) wrote that "Appropriate attention to altruism...substantially changes the theory of the firm, and consequently, theories of the economy." This article is one of the first to extend altruism to the economically important domain of the family firm, and to provide empirical evidence that lends support to this extension. While the use of secondary sources limits the generalizability of this study, Ilgen (1986) and others (Sackett and Larson, 1990, p. 435) point out that representativeness is less of a concern when the research question concerns the circumstances in which a particular effect can occur, as opposed to determining the frequency or relative strength of the observed effects. We are therefore encouraged that the results appear to identify circumstances in which altruism moderates the influence of pay incentives on the performance of this set of family firms. Consequently, and contrary to the prediction of agency models that do not account for altruism, family ownership does not appear to represent the kind of governance panacea that Fama and Jensen (1983) and others attribute to family owner-management. Rather, and consistent with the case-based studies about family firms, family-owned and -managed firms appear to experience agency problems that are costly to mitigate (Donovan, 1995; Levinson, 1971).

Interestingly, family firm CEOs appear to understand, or at least intuit, this need. Consistent with surveys of compensation practices in family firms, we found that 71% of the family firms included in our sample offer pay incentives to their family agents. This suggests that CEOs recognize the need to motivate family agents and attempt to shift additional risk onto the agent in the form of incentive compensation. Pay incentives, by themselves, however, do not appear to be sufficient. Rather, and consistent with our views, the data identify circumstances when their effectiveness appears to be contingent on information about the principal shareholders' estate and share transfer intentions and on whether the firm will remain under family control.

While it is our position that many family firm CEOs do indeed understand the need to offer incentives to family agents, it is also possible that they may find it necessary to offer incentives to nonfamily agents and extend the same to family agents to maintain equity. It is also possible that CEOs may offer both family and nonfamily agents pay incentives in an effort to signal their professionalism to prospective employees and other constituencies. Whatever the motive, we think it intriguing that altruism appears to explain why or when these pay incentives are effective, while the propositions based on professionalism and/or equity do not explain the responses observed in Fig. 2.

A rival explanation is that the agency problems experienced in family firms are rooted in free riding and not shirking or other forms of opportunistic behavior. The indicated solution to free riding, however, is not incentive compensation, but rather a flat wage. Interestingly, the relationships illustrated in Figs. 1 and 2 lend some support to this thesis. They show, for example, that flat wages are effective for firms that are expected to remain in family hands and keep estate transfer intentions secret, and for those firms in which estate intentions are known but retirement is distant. Interestingly, these results are also consistent with our theory since they, too, illustrate that information conditions matter, i.e., that effectiveness of this compensation strategy varies with the salience of the transfer commitment. Clearly, more work is needed to sort out altruism's effects.

The usual caveats apply. While the use of these data—and the constraints we placed on our sample—limit the generalizability of our conclusions, the 1995 Arthur Andersen survey presented us with a unique opportunity to test our theory using data that are difficult and quite expensive to obtain. We also lack means to externally confirm the reliability of the data. Further, the accuracy of some of our estimates was reduced because some of the measures used in the Arthur Andersen survey were categorical, and therefore more coarse, than continuous measures. However, that coarseness lends a conservative bias to the analysis since categorical measures deflate the variance and the likelihood of obtaining significant results (Hunter and Schmidt, 1990). We are also encouraged that we found no departures from the theoretically predicted pattern of positive and null results, despite the quality of the measures and use of 11 covariates.

In conclusion, these results lead us to believe that the economic literature on altruism is a potentially useful resource that promises to inform practice, suggest new directions for family business research, and lead researchers toward a richer theory of the family firm.

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We would like to thank Mike Carney and Eric Gedajlovic and colleagues at Case Western and the University of Connecticut for their comments on earlier drafts of this article. Survey data were provided courtesy of Arthur Andersen & Co. The *American Family Business Survey 1995* was conducted by the Arthur Andersen Center for Family Business in association with Loyola University Chicago Family Business Center and the Family Enterprise Center at Kennesaw State College. The support of the Family Business Program at the University of Connecticut and the H.R. Horvitz Family Foundation is gratefully acknowledged.

Appendix A

Ten covariates were included to reduce variance that is extraneous to the research question. We controlled for firm size, using the log transformation of total firm sales to correct for its skew, and for firm age, using a continuous measure calibrated in years. Differences in rates of sales growth associated with export sales are controlled using a five-level indicator ranging from 0 through >50%. Variance in performance due to technological intensity is controlled using a Likert-scaled item that asked, "How important are investments in information technology for the accomplishment of your future goals?" The potential influence of multiple family ownership on the CEO's ability to determine compensation practice is controlled with a dummy variable that identifies firms in which two or more families own at least 15% of the firm. We also used the number of employed family members and family ownership to control for the effect of ownership on compensation policy. Family ownership is the reported percentage of the firm's shares held by eight largest shareholders. Lastly, we used a dummy variable, dividend (coded 1 if the firm pays dividends) in an effort to assure that the reported annual or long-term cash bonuses are a contingent portion of salary and not dividend payments that are distributed in the form of a salary bonus with the intent of reducing the tax burden of family members.

We also controlled for capital intensity since the agency effects of debt (Fama, 1980) are likely to have greater salience in capital-intensive as opposed to non-capital-intensive industries. Since the industry categories identified in the survey do not correspond directly to SIC-based industry descriptions, capital intensity was coded 1 if the mean on the reported debt/equity ratio for the industry category was greater than the reported median debt/equity ratio for all industry categories, and 0 if below the median. Another dummy variable, industry growth, controls for variation in sales growth across the industries represented in the sample and is coded in the same manner. While coarse-grained, the resulting measures distinguishes capital intensive industries (e.g., manufacturing and transportation) from those that are not (e.g., retail) as well as industries that enjoyed high levels of sales growth during this period (e.g., telecommunications).

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